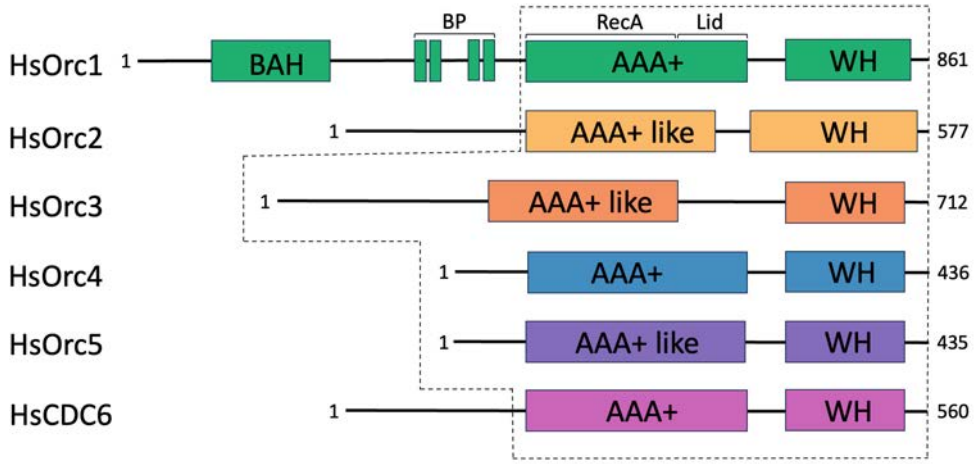
Figure 1 Supplement

A



B

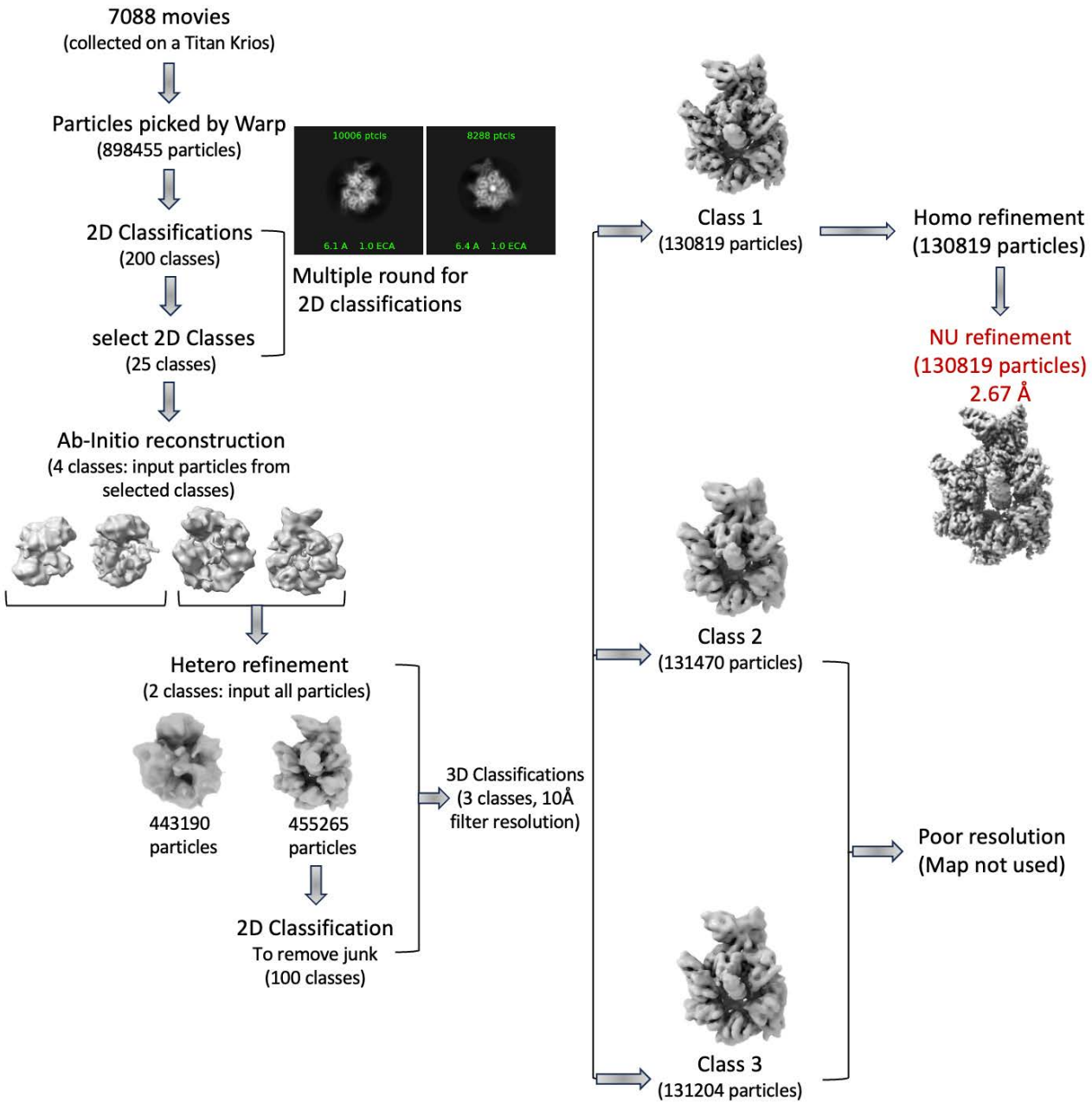
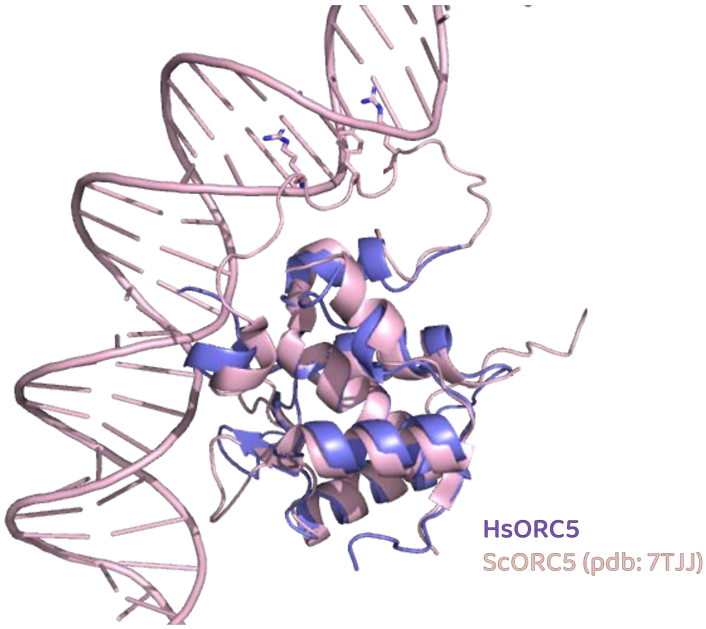


Figure 1 Supplement (Cont'd)

C



D

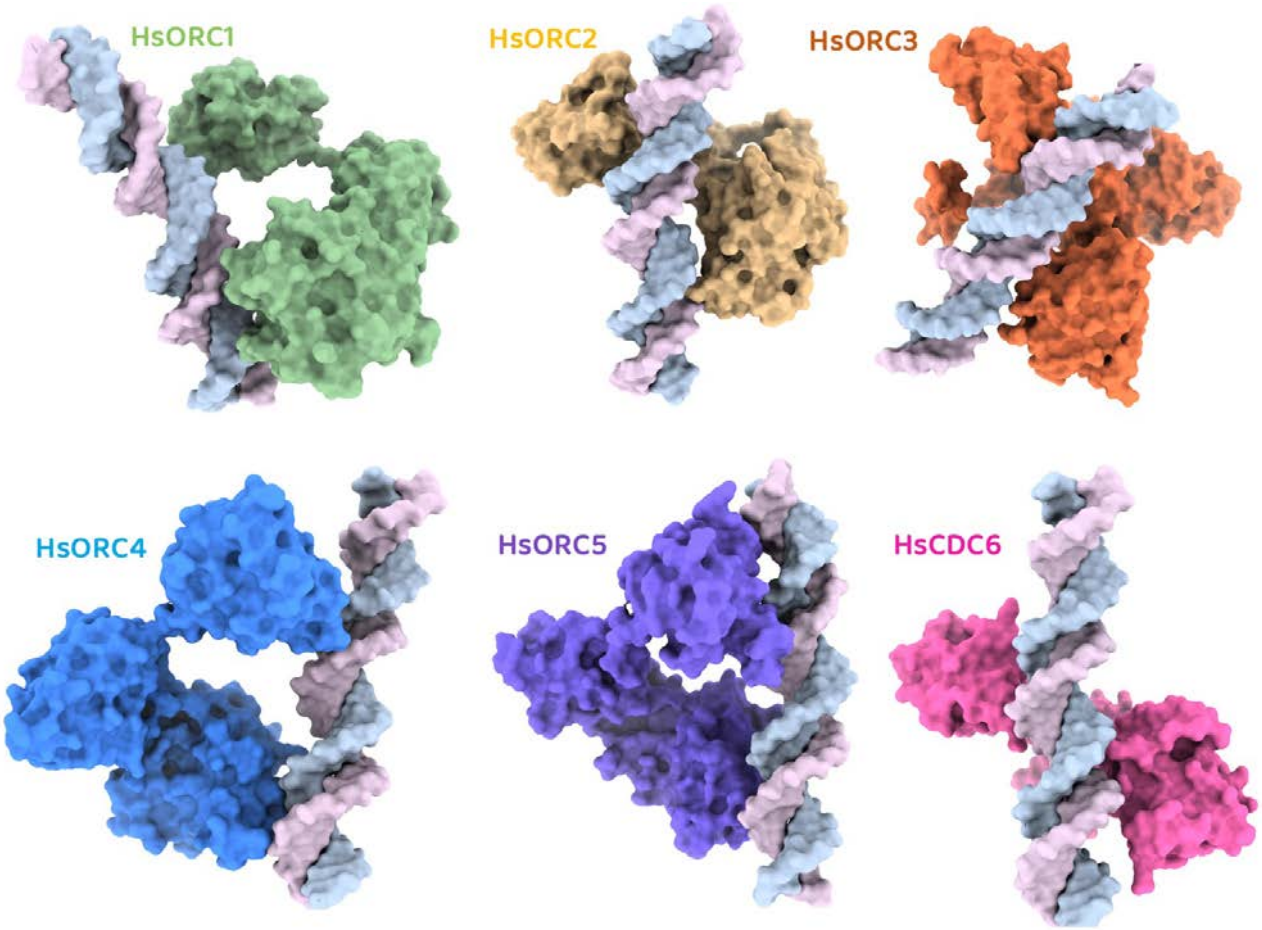
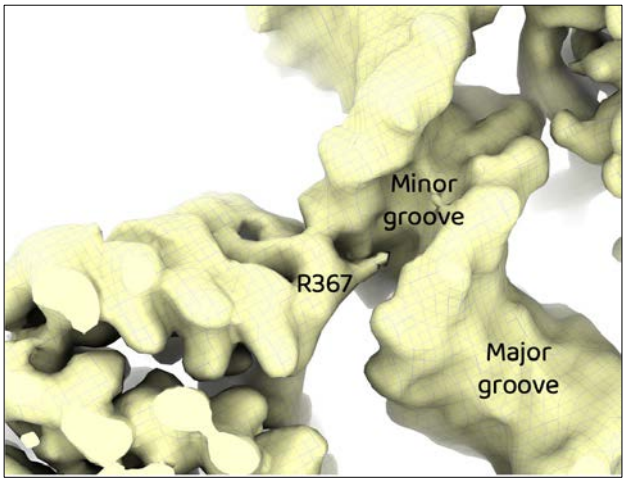


Figure 1 Supplement (Cont'd)

E



F

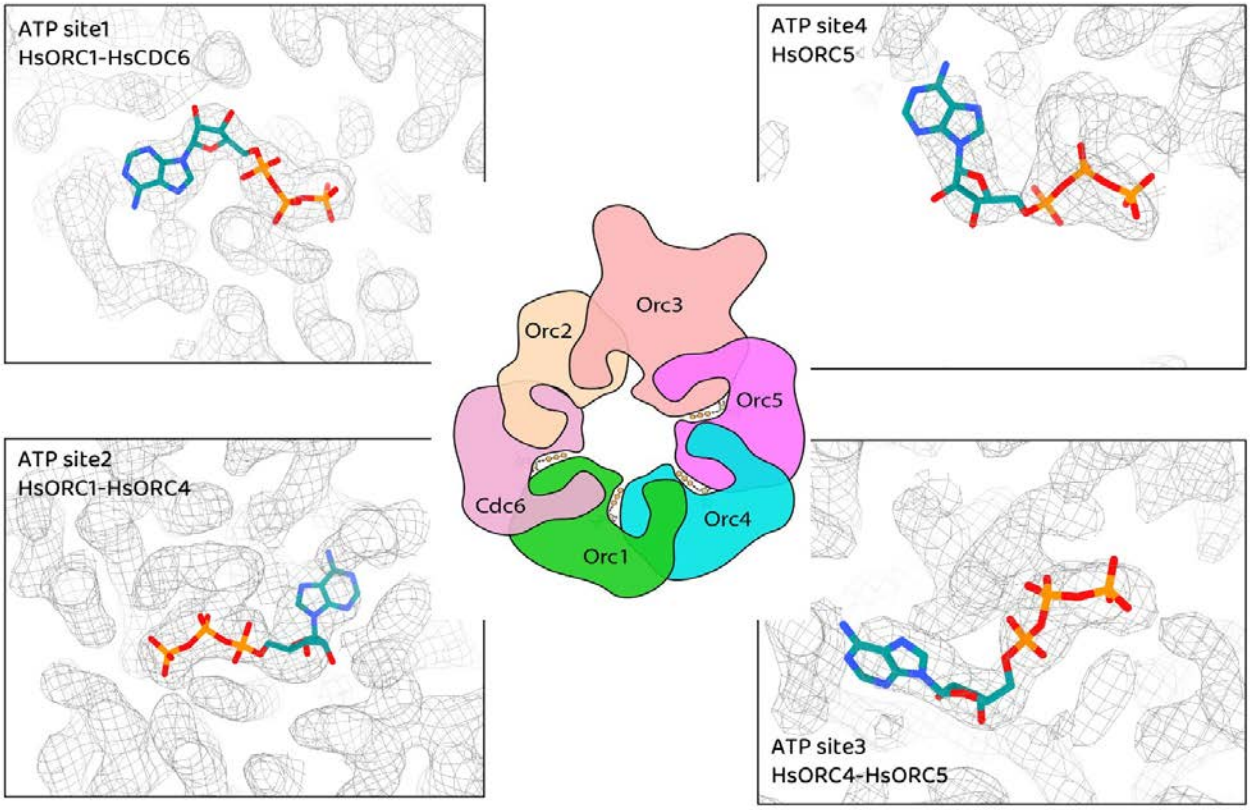


Figure S1 (related to Figure 1). (A) Domain organization and subunit–DNA interactions in the HsORC–Cdc6 complex. Schematic representation of the domain organization of all HsORC subunits (Orc1–5) and HsCdc6. Protein regions that were resolved and modeled in the cryo-EM structure are outlined with dashed gray lines. **(B) Cryo-EM workflow of HsODC structure.** Motion correction, averaging, CTF estimation, and particle picking steps were performed on movies in real time via the WARP (v1.0.9) software package. To determine an *ab initio* reconstruction, 2D classification and all further processing steps were performed in Cryosparc v3.2. See Methods section for details. **(C) Structure alignment of the human ORC5 subunit with the *S. cerevisiae* DNA-bound ORC5.** Alignment shows that *S. cerevisiae* ORC5 contains an extended, flexible loop enriched in basic residues that engages DNA and supports pronounced DNA bending, whereas the corresponding region in human ORC5 is disordered, resulting in limited DNA contact in the human complex. **(D)** Structural views of individual protein subunits interacting with DNA, derived from the cryo-EM density map. Figures were generated using ChimeraX. **(E)** Close-up view of the density map around residue R367 from the HsORC2 subunit. R367 is observed making direct contacts with DNA bases within the minor groove. **(F)** Cryo-EM density surrounding the ATP molecules at all four conserved ATP-binding sites within the HsORC–Cdc6 complex. The density is well-resolved, allowing precise placement of ATP and modeling of coordinating residues.

Figure 2 Supplement

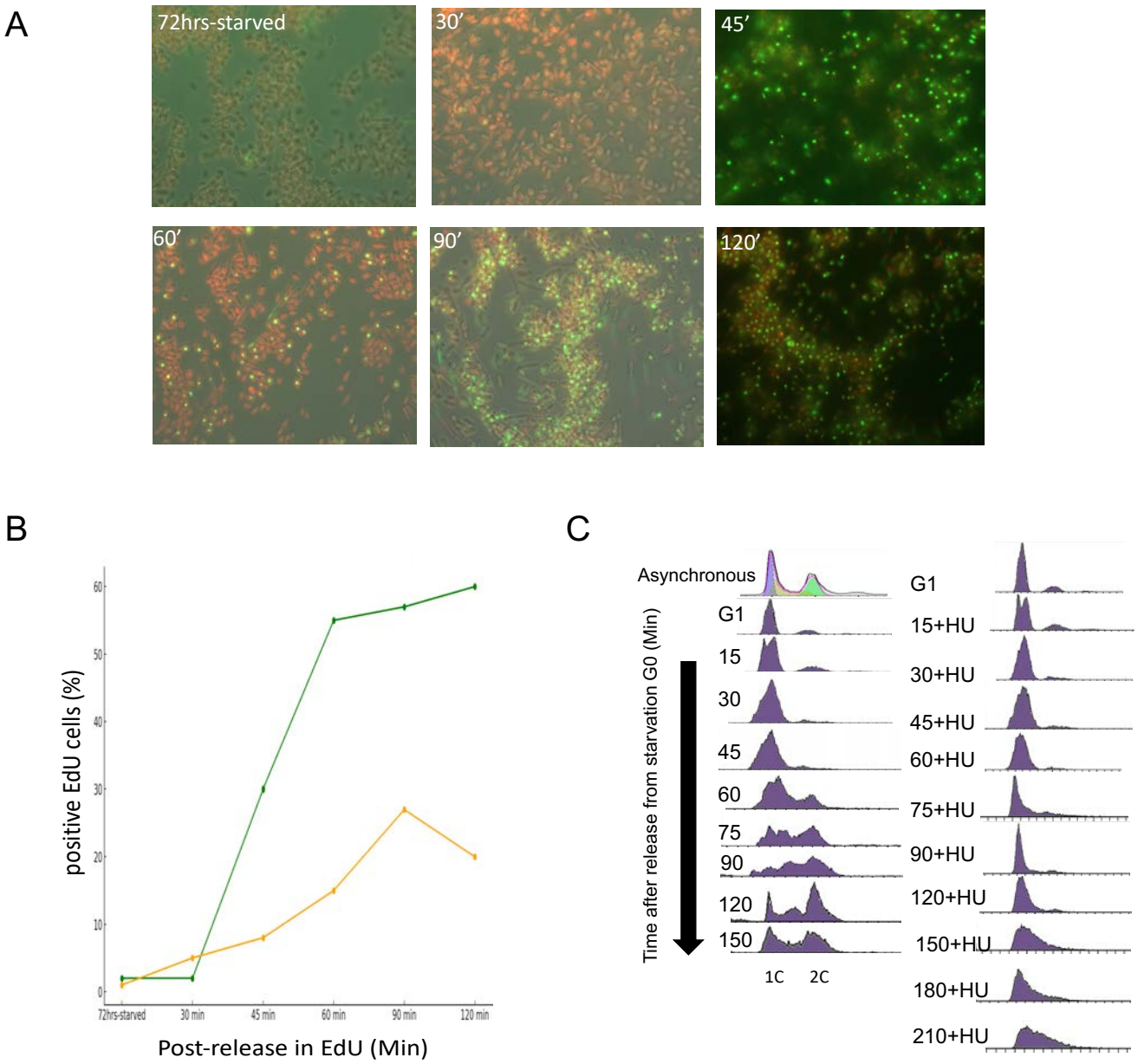


Figure S2 (related to Figure 2). Time-Course Analysis of S-Phase Progression in the Tk+ Strain Using EdU Incorporation. (A) Fluorescence microscopy images cells either starved for released into fresh, complete media for various times were treated with 100 μM EdU and DNA synthesis was visualized using Click-iT™ chemistry with Alexa Fluor™ 488 (green fluorescence). Red fluorescence (DAPI) was used to stain total nuclear DNA for cell visualization. **(B)** The percentage of EdU-positive cells at each time point (Green line) and percentage of budded cells (orange line). **(C)** flow cytometry of DNA content in asynchronously growing cells, cells arrested in G1 by starvation fro 72 hours and cells released for different times in the absence (left) or presence (right) of 5 mM HU.

Figure 3 Supplement

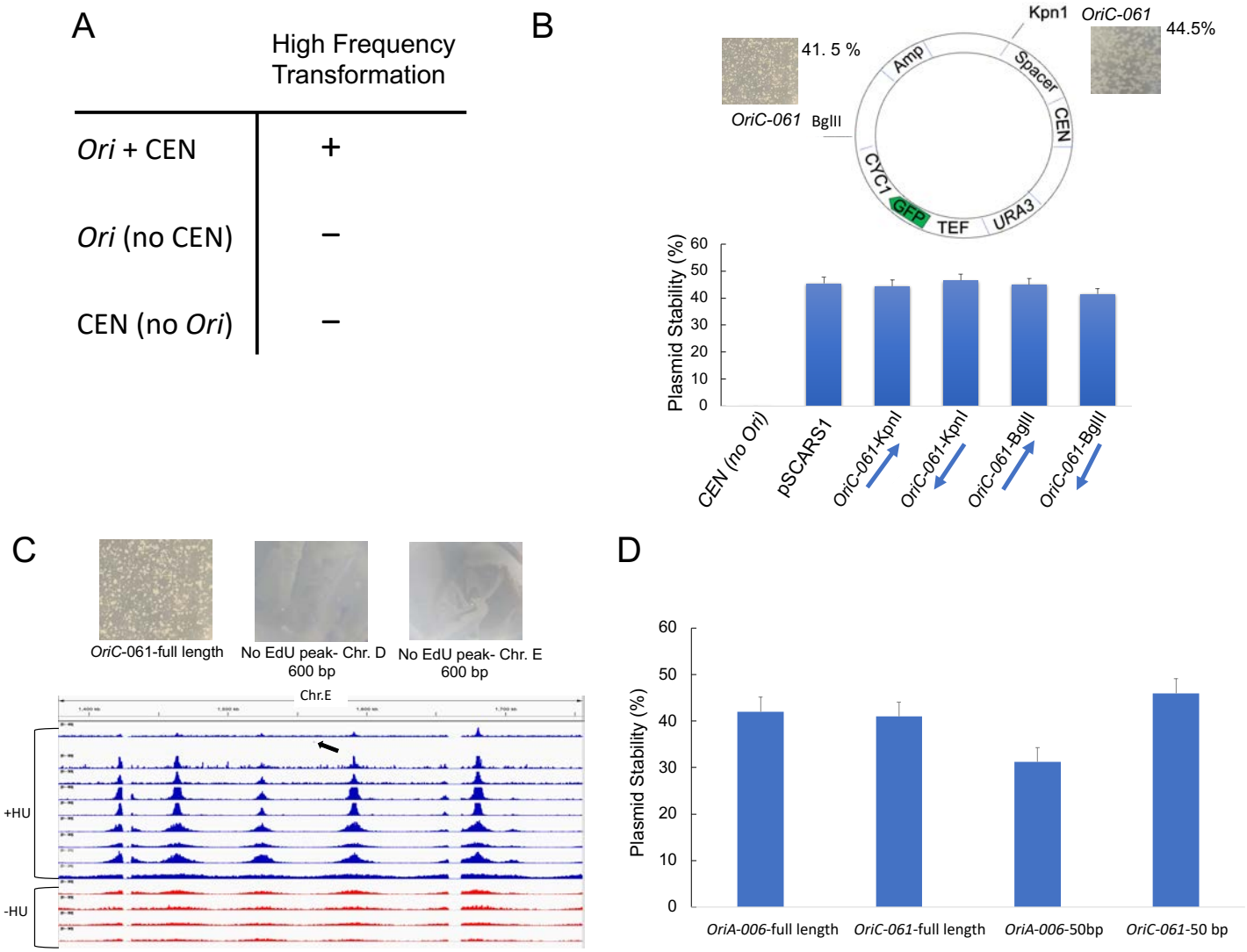


Figure S3 (related to Figure 3). Plasmid assay for origin activity. (A) in *Y. lipolytica*, autonomous replication of a plasmid required both a centromere (*CEN*) and origin of DNA replication (*Ori)* Vernis et al. DOI: 10.1128/mcb.17.4.1995. The plasmid has the *URA3* selectable marker. **(B)** Plasmid construct showing a CEN, and two *Ori* insertions sites at either Kpn1 of BglII. When *OriC-061* was inserted at either site, high frequency transformation occurred and the plasmid stability was comparable. **(C)** The *OriC-061* origin, but not a 600 bp form Chromosome D or Chromosome E (arrow showing region between two origins) show high frequency transformation upon *URA3* selection. The *OriC-061* plasmid is the same as shown in panel B. **(D)** 50 base pair region containing the core 30 base pair region of *OriA-006* or *OriC-061* function as efficiently as a full length 600 base pair region of the genome. The graph shows percentage of plasmid stability and bars represent mean ± SD from three independent replicates.

Figure 4 Supplement

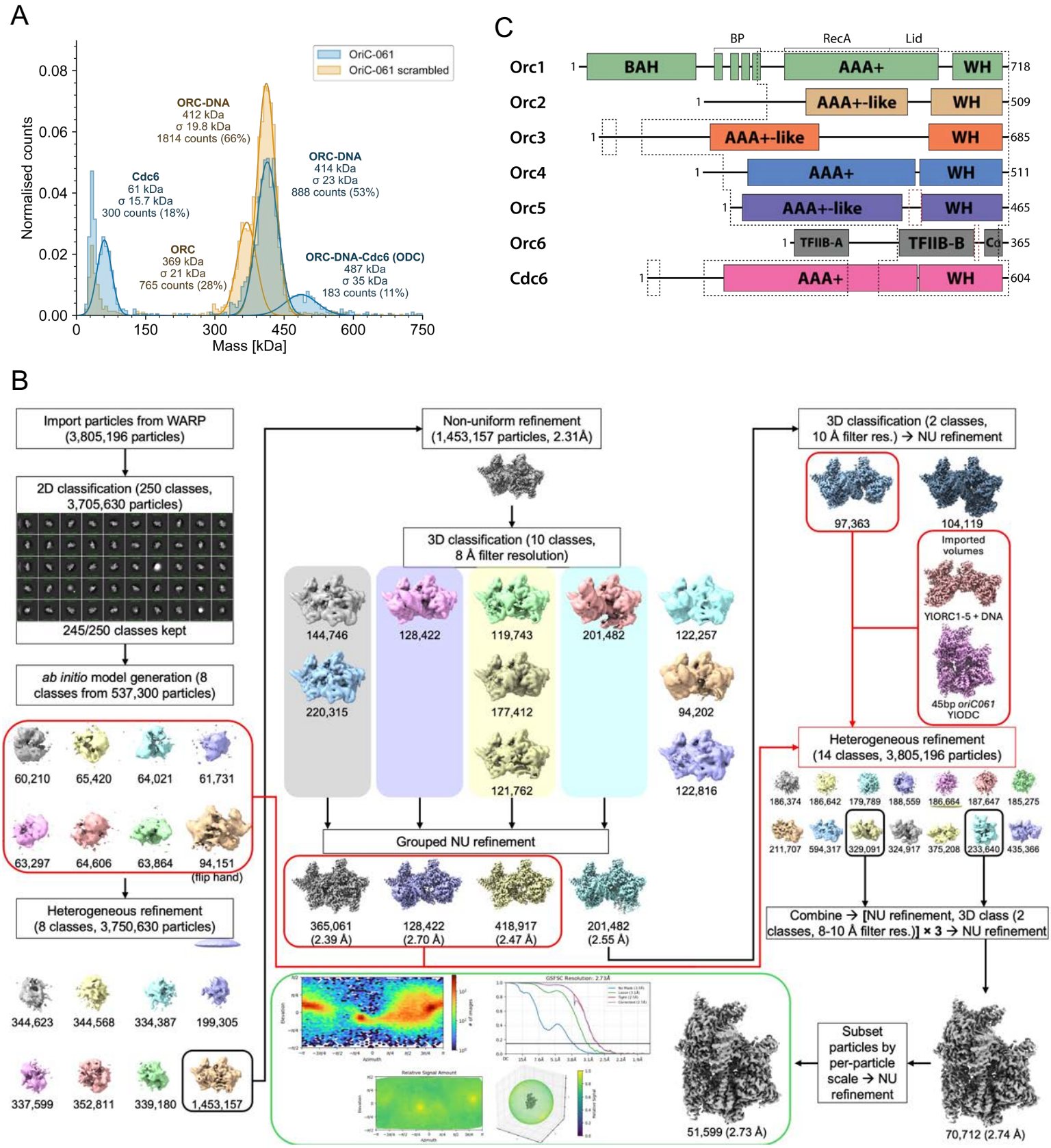
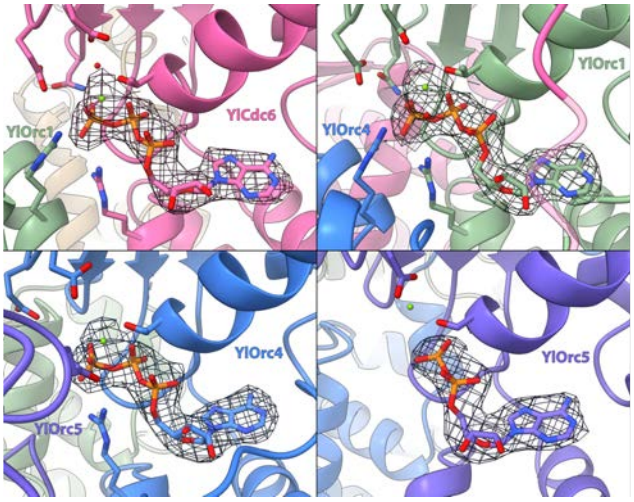
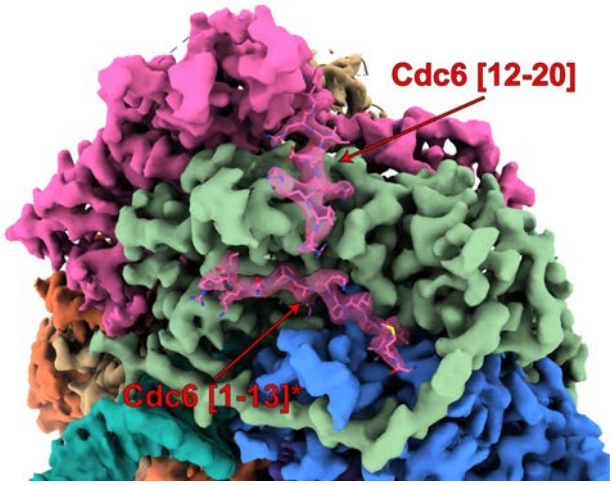


Figure 4 Supplement (Cont'd)

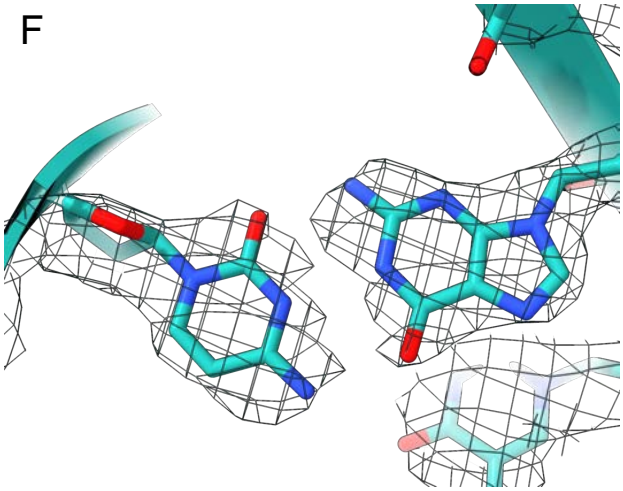
D



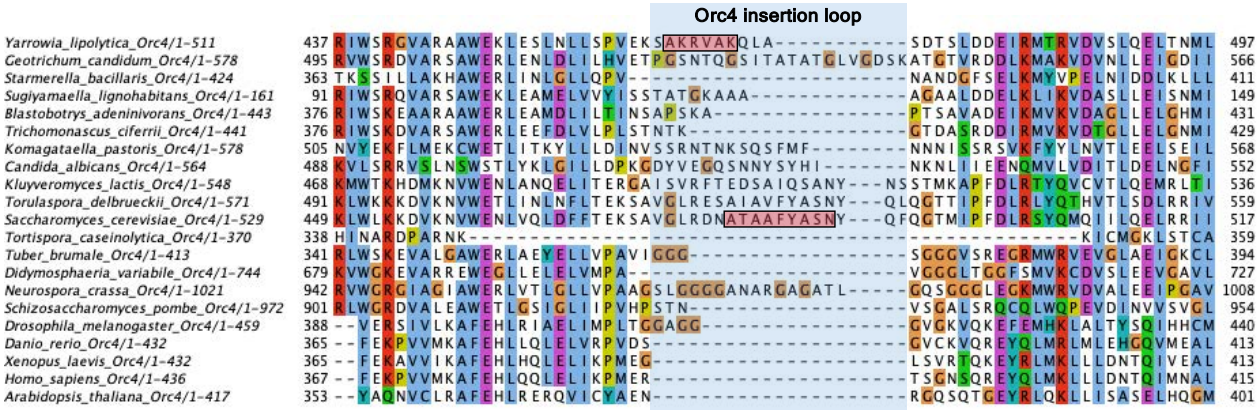
E



F



G



H

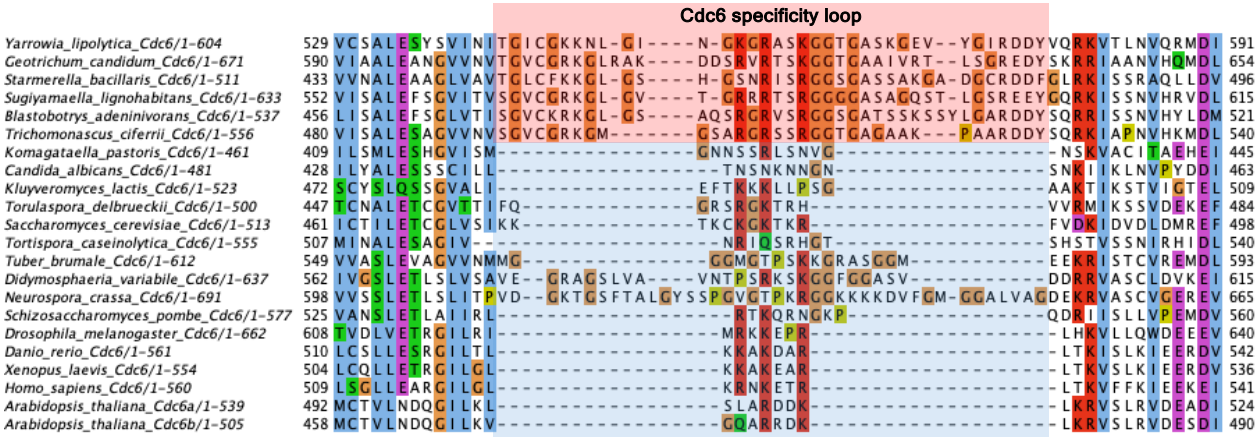


Figure 4 Supplement (Cont'd)

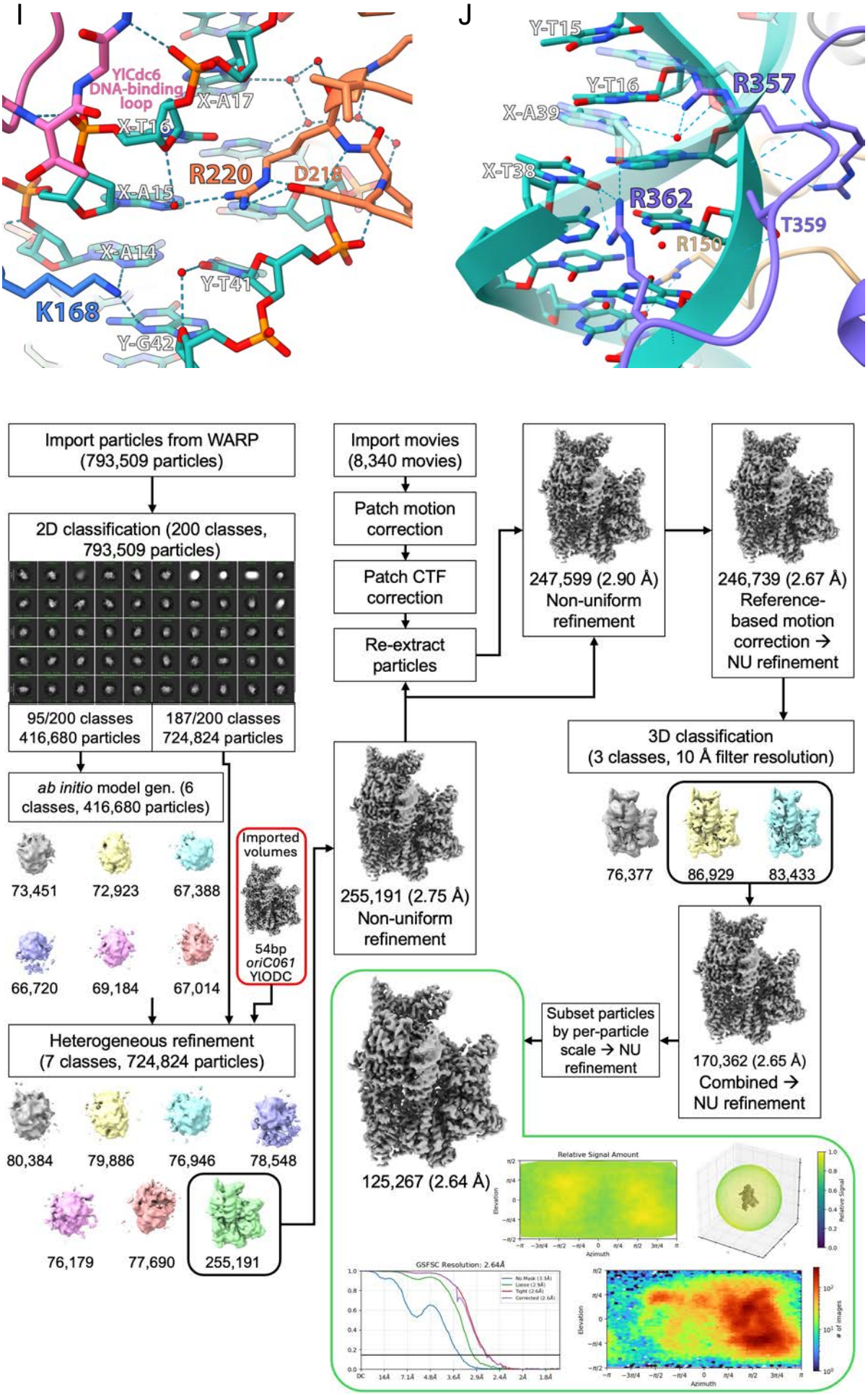
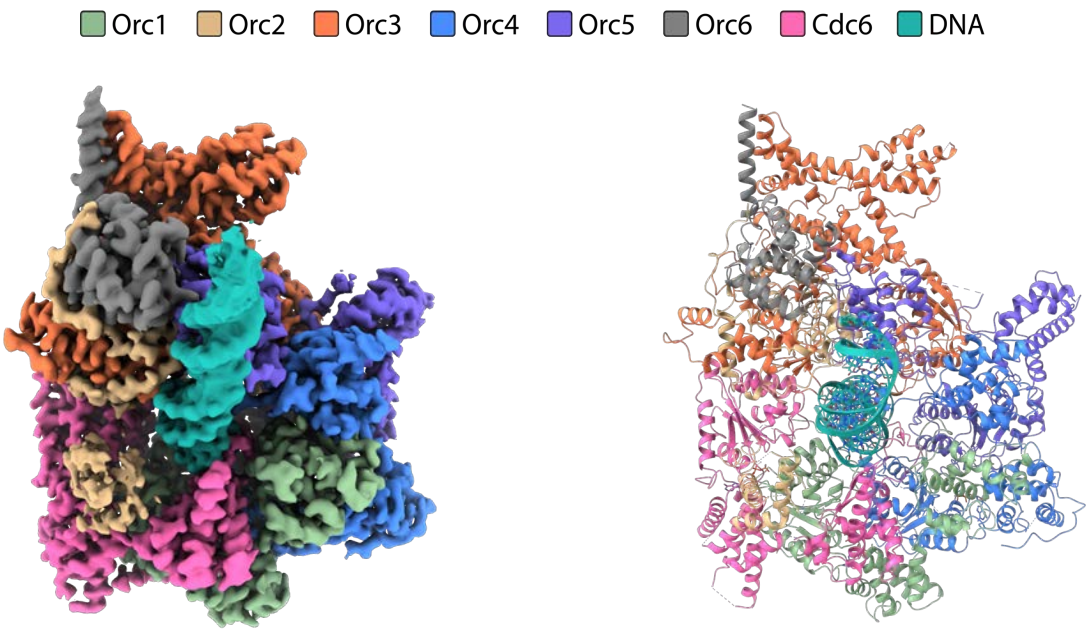
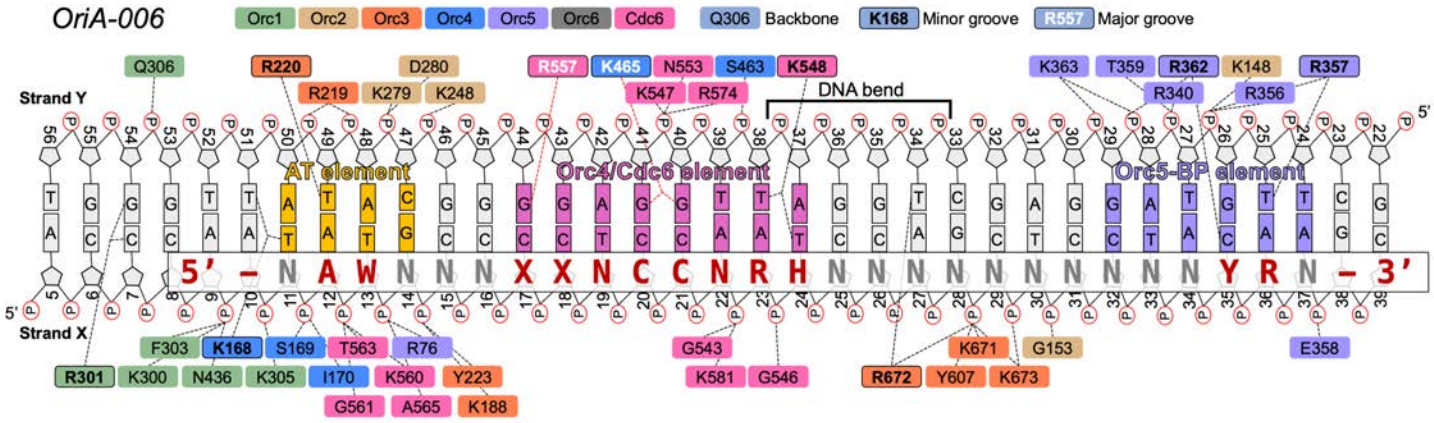


Figure 4 Supplement (Cont'd)

L



M



N

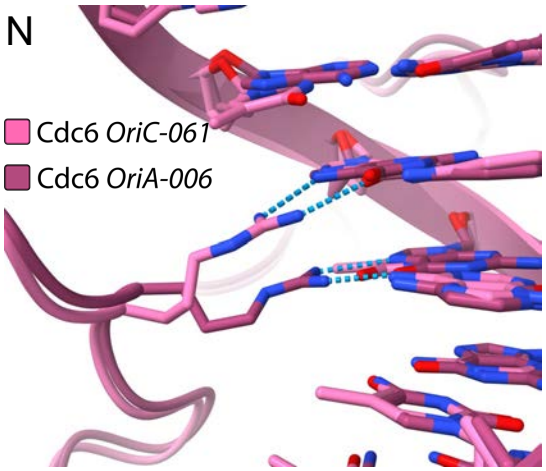


Figure S4 (related to Figure 4). (A) Mass photometry data of the gel filtration peak fractions from the loading assay in Figure 4A. **(B)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YlORC-DNA$^{54bpOriC-061}$-YlCdc6 cryo-EM map. **(C)** Schematic of the domain organization of YlORC and YlCdc6. Dotted lines surround the regions visible within either YlORC-DNA-YlCdc6 maps.

Figure S4 (continued). (D) ADP/ATP in stick representation at the four ATP-binding sites within the YlORC-DNA$^{54bpOriC-061}$-YlCdc6 complex, with a mesh overlay of the representative cryo-EM density of these nucleotides. **(E)** A view of the YlCdc6 IDR sections binding to the YlOrc1•YlOrc4 AAA domain interface and the YlOrc1-AAA•YlCdc6-WHD interface. * denotes the visibility of a leftover glycine on the N-terminus of YlCdc6 resulting from TEV protease cleavage of an N-terminal 8xHis-SUMO tag. **(F)** Representative cryo-EM density of a base pair of *OriC-061* in the YlORC-DNA$^{54bpOriC-061}$-YlCdc6 complex. **(G)** An MSA of the Orc4 region surrounding the Orc4 insertion loop including various related yeasts and eukaryotes. Pink boxes in the YlOrc4 and ScOrc4 sequences denote the region that encompasses the species' insertion helix. **(H)** An MSA of the Cdc6-WHD surrounding the specificity loop. The red box indicates species containing a conserved specificity loop, while blue indicates those that appear to lack sequence homology in this region.

Figure S4 (continued). (I) View of the AT element (positions X-A14 to X-A17) of *OriC-061* in the YlORC-DNA$^{54bpOriC-061}$-YlCdc6 structure, displaying the hydrogen bond interactions between protein, DNA, and water. **(J)** Same as **(I)** but of the Orc5 basic patch (Orc5-BP) element (positions X-G35 to X-A40) of *OriC-061*. **(K)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YlORC-DNA$^{60bpOriA-006}$-YlCdc6 cryo-EM map.

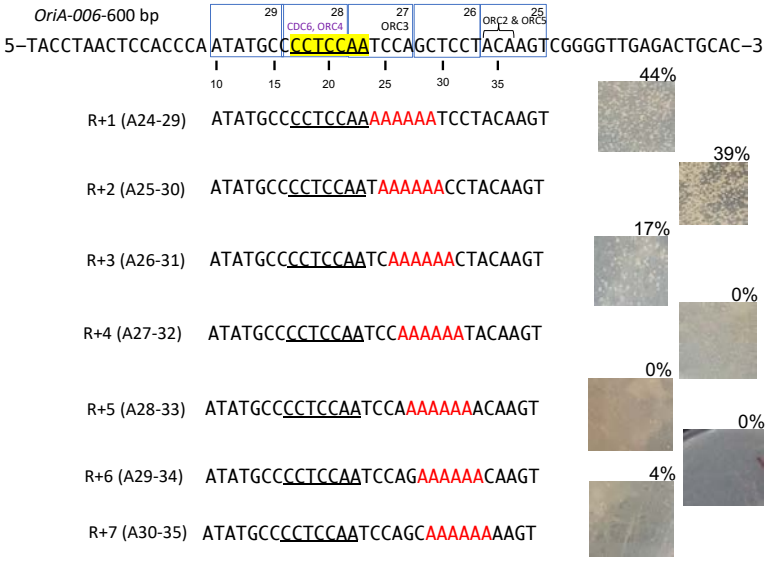
Figure S4 (continued). (L, M) Same as Figure 4B and 4C, respectively, but of the 2.6 Å resolution cryo-EM map YlORC-DNA$^{60bpOriA-006}$-YlCdc6 and corresponding structure. **(N)** A view of the difference in the major groove base interaction of YlCdc6 R557 between YlORC-DNA$^{54bpOriC-061}$-YlCdc6 and YlORC-DNA$^{60bpOriA-006}$-YlCdc6.

Figure 5 Supplement Figures

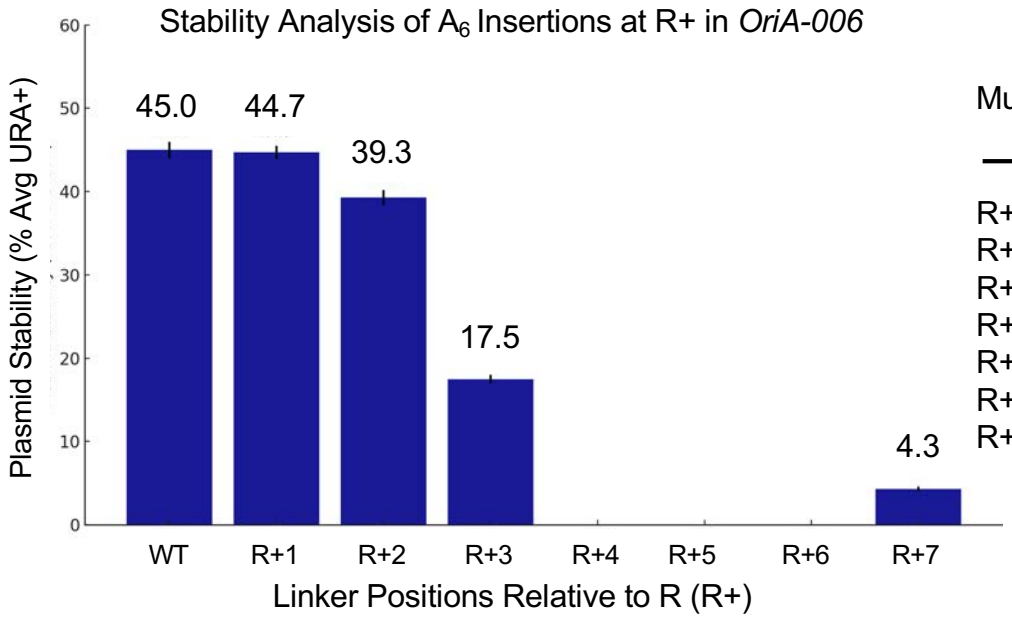
A

Origin	Mutant	Orc4-Cdc6 Motif *OriC* CCNCCNR/ *OriA* CNNCCNR	ARS activity	Plasmid Stability (%)	Cdc6 Loading Efficiency (%)
OriC	WT	CACCCA	+++	47	100
OriC	C1A	ANCCNA	+++	48.6	NT
OriC	C2A	CNACNA	+++	47.5	NT
OriC	C3A	CNCANA	+++	48.8	NT
OriC	C1,C2,C3	GNGGNA	–	0	NT
OriC	C1,C2,C3	TNTTNA	–	0	NT
OriC	R to C	CACCCC	++	4	NT
OriA	WT	CCTCCAA	+++	47	NT
OriA	C1A	ANNCCNA	+++	48	NT
OriA	C2A	CNNCANA	+++	47.3	NT
OriA	C1G	GNNCCNA	+++	47	105.1 ± 1.3
OriA	C2C3G	CNNGGNA	+++	22	88.0 ± 2.0
OriA	C1C2C3	GNNGGNA	–	0	46.9 ± 0.7
OriA	C1C2C3	ANNAANA	–	0	20.3 ± 0.8
OriA	C1C2C3	TNNTTNA	–	0	NT
OriA	R to C	CCTCCAC	++	4	NT

B



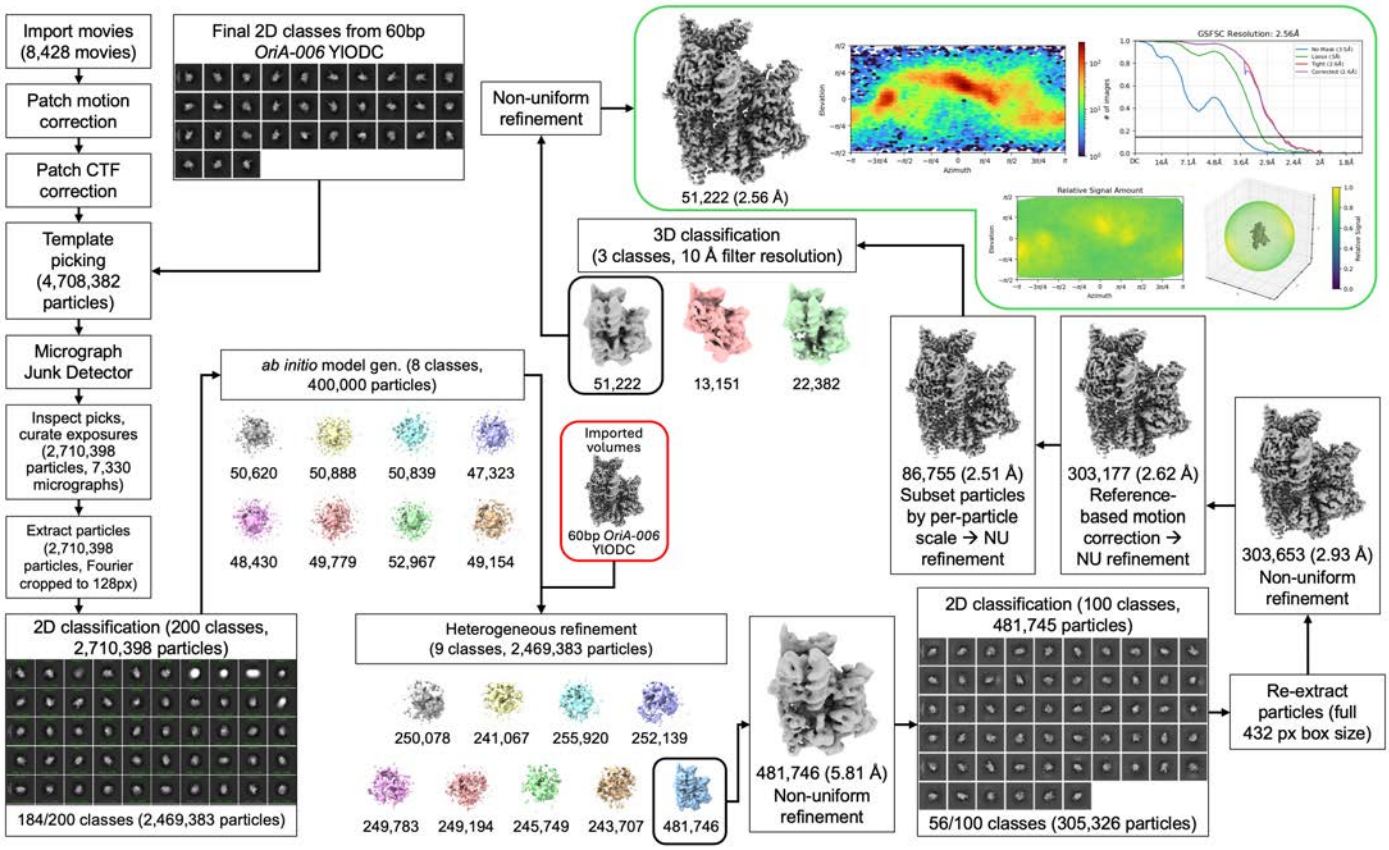
C



Mutant	Normalized Cdc6 loading %
R+1	43.6 ± 1.5
R+2	86.1 ± 3.2
R+3	90.4 ± 2.7
R+4	90.4 ±6.5
R+5	0
R+6	0
R+7	0

Figure 5 Supplement (Cont'd)

D



E

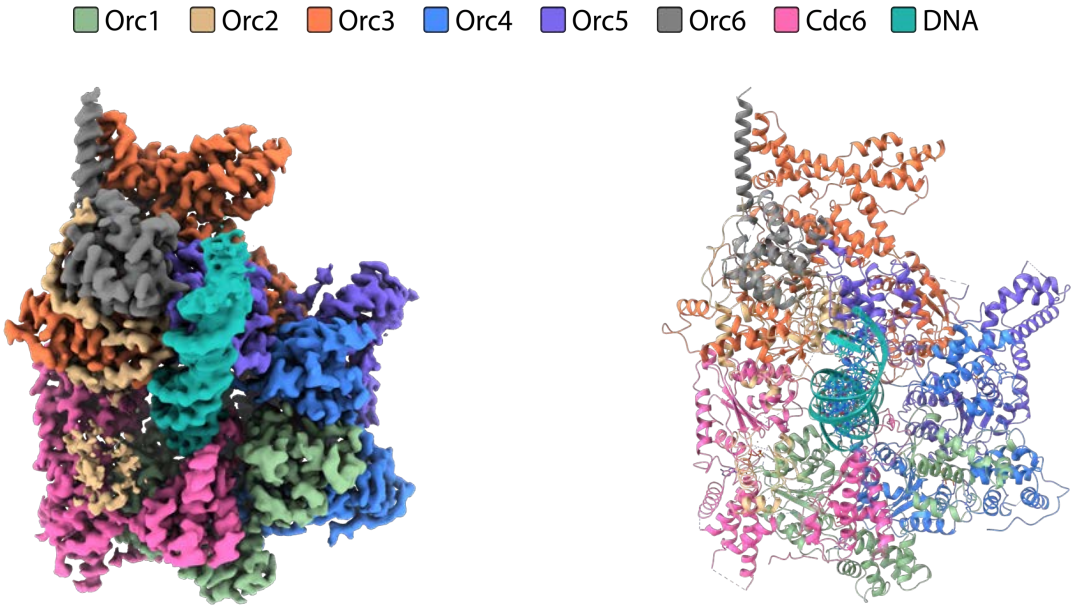


Figure S5 (related to Figure 5). Analysis of *Y. lipolytica* mutant origins. (A) Summary table of mutations in *Oria-006* and *OriC-061 reporting* high frequency transformation *(ARS* activity*)* of point mutations and A_6 insertions. **(B)** Analysis of mutant *OriA-006* with insertion of six A residues at different positions. The high frequency transformation images and percent plasmid stability was done as described in Figure 5 legend. **(C)** Summary of the plasmid stability of the A_6 insertion mutations. **(D)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YlORC-DNA$^{60bp\mathit{OriA\text{-}CNNGGNR}}$-YlCdc6 cryo-EM map. **(E)** Same as Figure 4B and Figure S4L, but of the 2.6 Å resolution YlORC-DNA$^{60bpOriA\text{-}CNNGGNR}$-YlCdc6 cryo-EM map and the accompanying structure.

Figure 6 Supplement

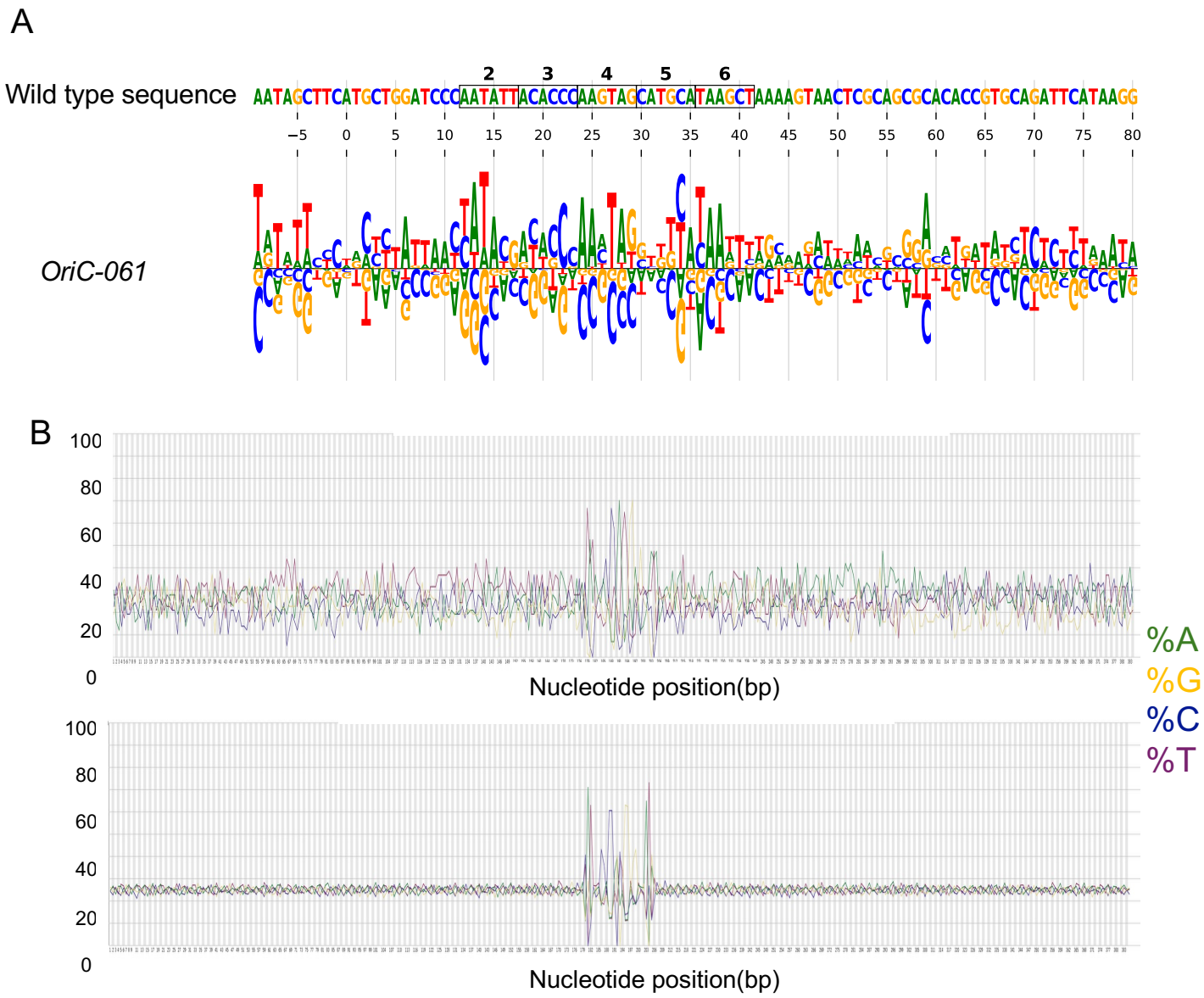


Figure S6 (related to Figure 6). (A) Additive model for origin specificity derived from an MPOS assay on 90 bp sequences containing *OriC-061*. Boxed region indicates the core region identified by linker scanning (linker positions 2-6 in **Fig. 3A**). Note that the base numbers refer to the position within the 90 bp and the numbers refer to the numbering of nucleotides in Figure 4C. **(B).** Analysis of nucleotide frequencies across 54 *Y. lipolytica early* origins of replication compared to 5351 non-origin sequences matched the consensus sequence. The early origins showed a skew for T/A sequences 5' to the core consensus sequence and a skew for A/T base pairs 3' to the core consensus sequence. There is no skew surrounding the non-origin consensus sequences. There are 5545 matches to the consensus sequence in the genome.